

Fake News Detection System Using Python and Tkinter

A Rule-Based Keyword Detection Approach with GUI Implementation

Course Code: CSA0801

Course Name: Python Programming

Team Members

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Introduction

The Problem

Fake news spreads rapidly through websites and social media — misleading headlines and sensational claims make false information difficult to identify.

The Challenge

Manually checking every piece of information is time-consuming and impractical in today's fast-moving information environment.

Our Solution

A simple Python desktop application that analyzes news text using predefined suspicious keywords, providing quick preliminary classification through an easy-to-use GUI.



Objectives



1 Build a Desktop GUI

Develop a simple, user-friendly interface using Python Tkinter with a ScrolledText input area.

2 Detect Suspicious Content

Identify and count suspicious keywords and phrases present in any pasted news article.

3 Classify News Articles

Classify results as **Possibly Real**, **Suspicious**, or **Fake** based on keyword score.

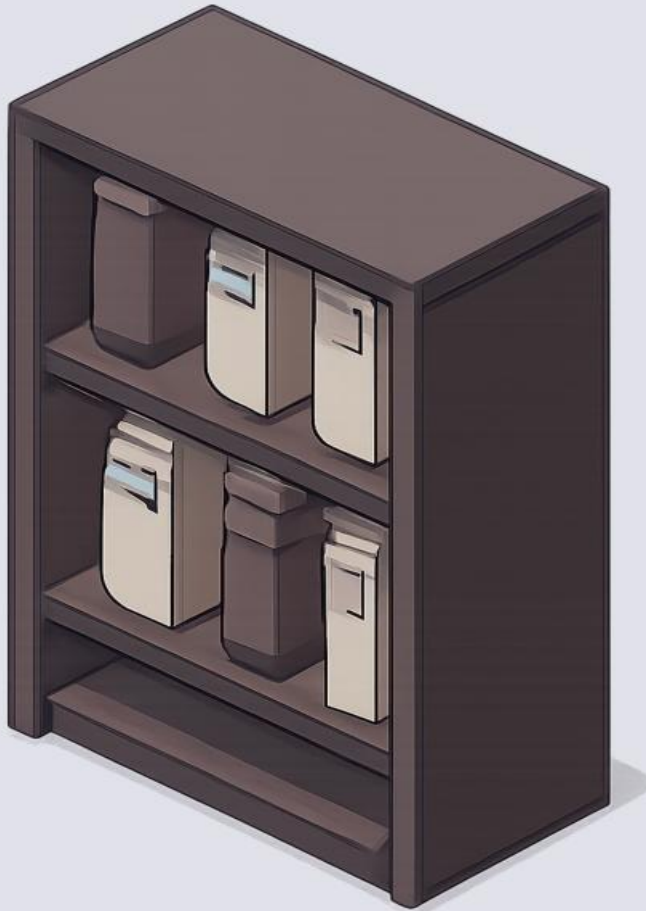
4 Demonstrate Python Concepts

Apply functions, lists, loops, strings, conditionals, and GUI programming in a real-world project.

Literature Survey

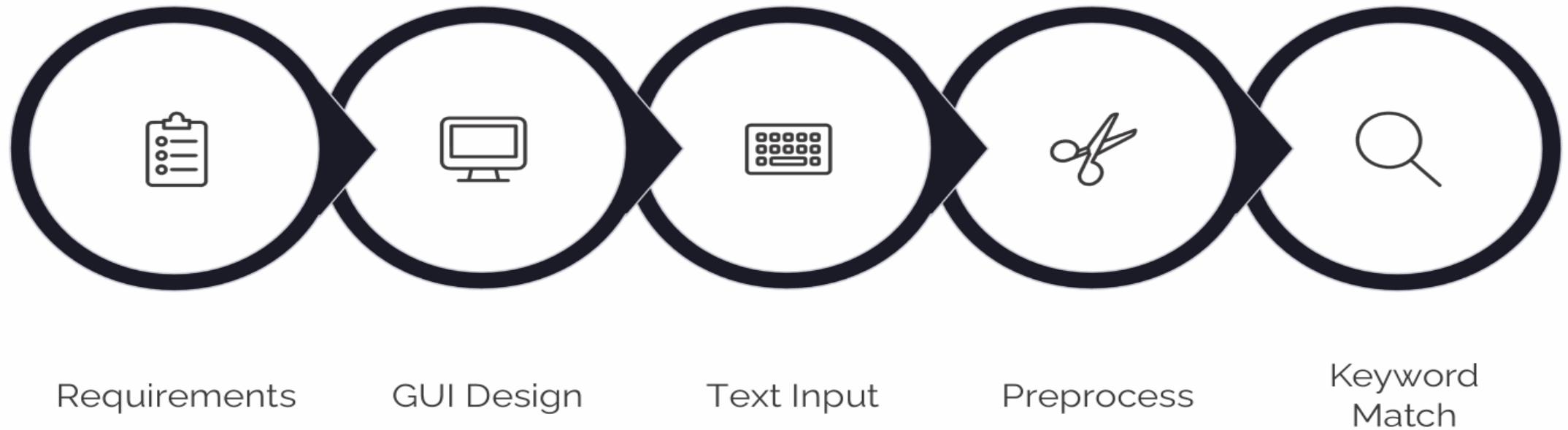
Recent research on fake news detection (2024-2026) demonstrates the growing importance of automated classification systems:

Author(s)	Advantages	Disadvantages	Technologies Used
Sharma & Patel (2024)	Deep learning for misinformation detection; highly accurate; contextual understanding	Requires large labeled datasets; computationally expensive	Deep Learning, Neural Networks, TensorFlow
Chen et al. (2024)	Rule-based keyword detection; fast, transparent, beginner-friendly	Limited context understanding; false positives	Python, String Matching, Lists
Rodriguez et al. (2024)	Comparative ML models analysis; multiple algorithm comparison; accuracy metrics	Model selection complexity; training time	Random Forest, SVM, Logistic Regression, scikit-learn
Nakamura et al. (2025)	NLP for sensational language; detects emotional language patterns	Language-specific; requires NLP expertise	NLTK, spaCy, Word Embeddings
Thompson et al. (2025)	GUI-based verification systems; user-friendly interface; real-time feedback	Limited to predefined rules; scalability issues	Tkinter, Python, GUI Frameworks
Gupta et al. (2025)	Hybrid rule-based + ML approach; combines transparency with accuracy	Increased complexity; maintenance overhead	Python, scikit-learn, Custom Rules
Johnson et al. (2026)	Future directions in detection; contextual understanding; API integration	Still in research phase; not production-ready	Machine Learning, APIs, NLP, Databases

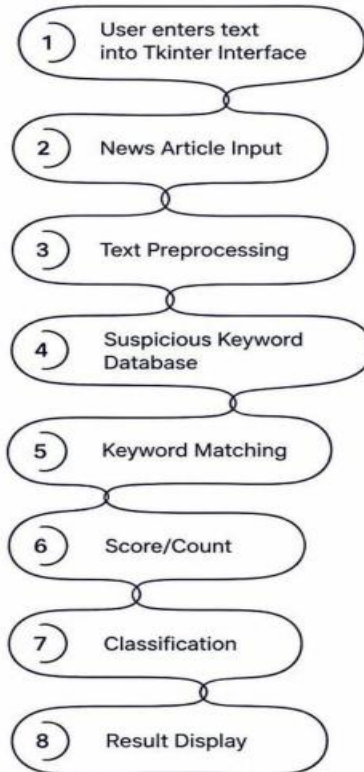


Methodology

The system follows a **rule-based keyword detection approach** — not machine learning — making it transparent, beginner-friendly, and fully explainable.



System Design



Three Possible Outputs

✓ POSSIBLY REAL

Low or zero suspicious keyword count

⚠ SUSPICIOUS

Moderate keyword matches detected

X FAKE NEWS

High count of suspicious phrases found

📌 The keyword database is the core intelligence of this rule-based system.

Module 1 — News Input & Text Processing

How It Works



User Input

User enters or pastes a news article into the Tkinter **ScrolledText** widget.



Validation

Empty input is detected and rejected immediately with an **orange warning** label.



Preprocessing

strip() removes leading/trailing spaces. **lower()** enables case-insensitive matching.

Code Snippet

```
def check_news():
    news = text_box.get(
        "1.0", tk.END
    ).strip().lower()

    if news == "":
        result_label.config(
            text="Please enter a news article!",
            fg="orange"
        )
    return
```

Sample Input

"The city administration announced a new public transportation project. Construction is expected to begin next month."

Output — Module 1



Fake News Detection System

FAKE NEWS DETECTION SYSTEM

Analyze news content using keyword-based detection

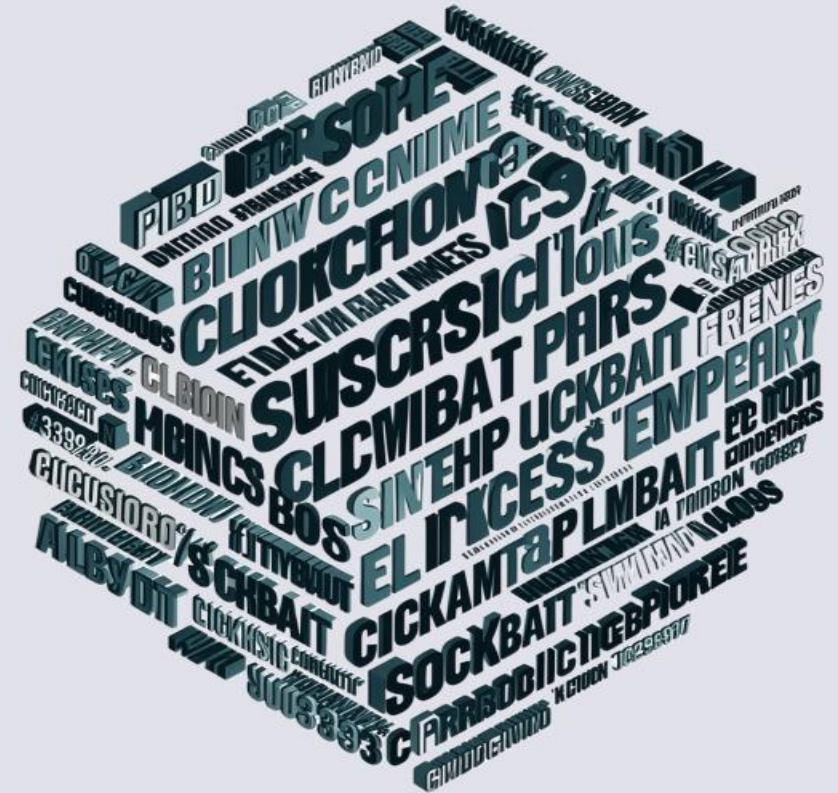
Enter News Article

CHECK NEWS

CLEAR

DETECTION RESULT

Please enter a news article!



Module 2 — Sample Output

Sample Input

"BREAKING! Scientists are shocked by this unbelievable miracle cure. Click here to get free money. Act now before it's too late!"

Keywords Detected

breaking

shocked

miracle

click here

free money

act now

Suspicious Keywords Found

6+

X Classification Result

The high keyword count triggers the **FAKE NEWS** classification — displayed prominently in the GUI result label.

Output — Module 2



Fake News Detection System

FAKE NEWS DETECTION SYSTEM

Analyze news content using keyword-based detection

Enter News Article

The health department announced a new vaccination program for children. Officials said the program will begin next month. However, social media posts are claiming that this is a secret cure and that doctors are hiding the truth from the public.

CHECK NEWS **CLEAR**

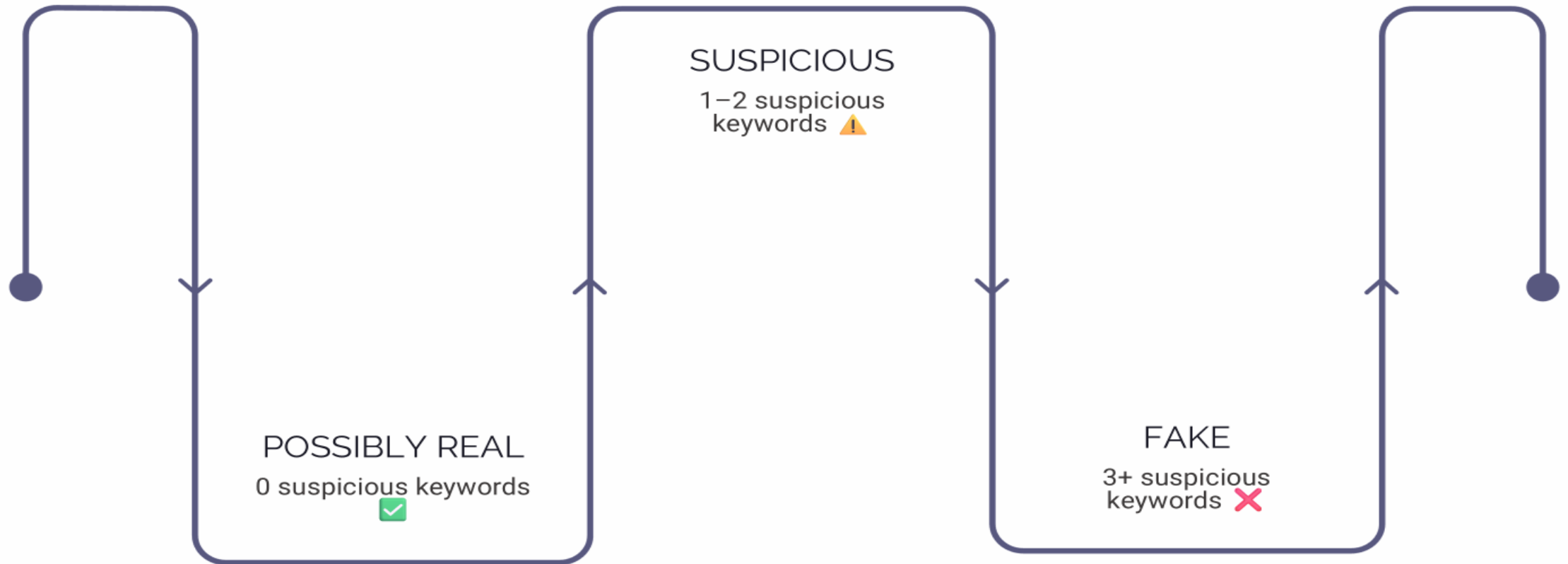
DETECTION RESULT

 **SUSPICIOUS NEWS**

Suspicious keywords found: 2

Module 3 — News Classification

The system classifies news articles based on the count of suspicious keywords found, following these predefined rules:



❏ This is a rule-based classification, not a guarantee that the article is factually fake.

Code — Module 3: News Classification

```
if keyword_count == 0:  
    result = "POSSIBLY REAL NEWS"  
elif keyword_count <= 2:  
    result = "SUSPICIOUS NEWS"  
else:  
    result = "FAKE NEWS"
```

```
score_label.config(  
    text=f"{result} ({keyword_count} keywords)",  
    fg="green" if keyword_count == 0 else "orange" if keyword_count <= 2 else "red"  
)
```

- Compares count against thresholds
- Displays appropriate message
- Changes label color
- Shows keyword count

Fake News Detection System

FAKE NEWS DETECTION SYSTEM

Analyze news content using keyword-based detection

Enter News Article

The city administration announced a new public transportation project. Construction is expected to begin next month, and officials said the project will improve connectivity between several residential areas.

CHECK NEWS

CLEAR

DETECTION RESULT

✓ POSSIBLY REAL NEWS

Suspicious keywords found: 0

Fake News Detection System

FAKE NEWS DETECTION SYSTEM

Analyze news content using keyword-based detection

Enter News Article

The health department announced a new vaccination program for children. Officials said the program will begin next month. However, social media posts are claiming that this is a secret cure and that doctors are hiding the truth from the public.

CHECK NEWS

CLEAR

DETECTION RESULT

⚠ SUSPICIOUS NEWS

Suspicious keywords found: 2

Fake News Detection System

FAKE NEWS DETECTION SYSTEM

Analyze news content using keyword-based detection

Enter News Article

You have been selected to receive free money. Claim your reward now and double your money instantly. This offer is 100% guaranteed with zero risk

CHECK NEWS

CLEAR

DETECTION RESULT

⚠ FAKE NEWS

Suspicious keywords found: 6

Implementation & Results



Tkinter GUI

Multi-line article input, a CHECK NEWS button, and a dynamic result label that updates based on classification.



Possibly Real

0–1 suspicious keywords detected. Displayed in green for likely factual, straightforward news.



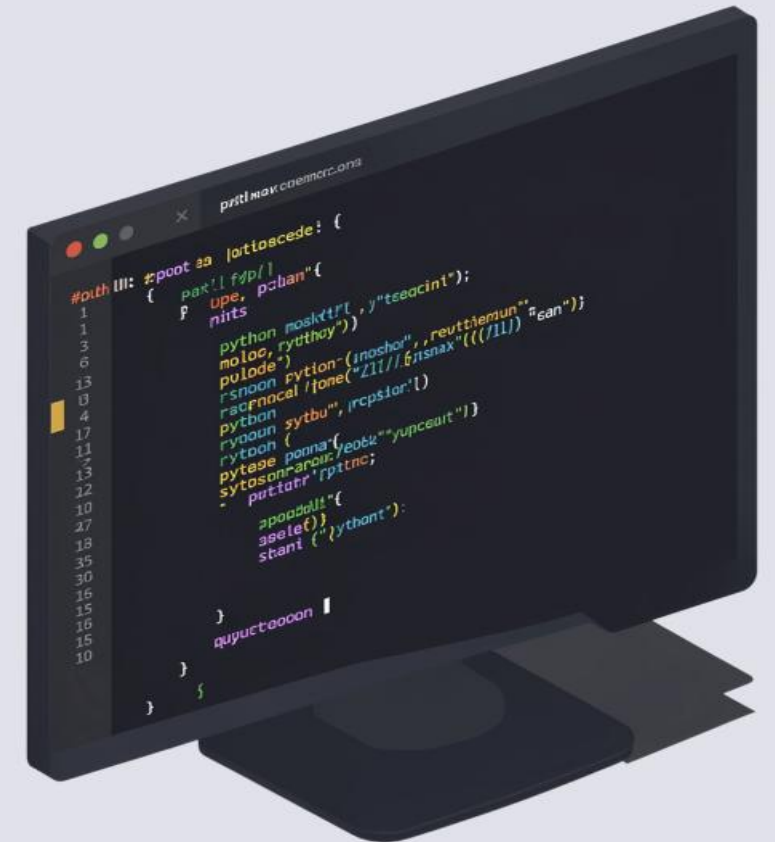
Suspicious

2–4 suspicious keywords detected. Displayed in orange when an article needs further verification.



Fake News

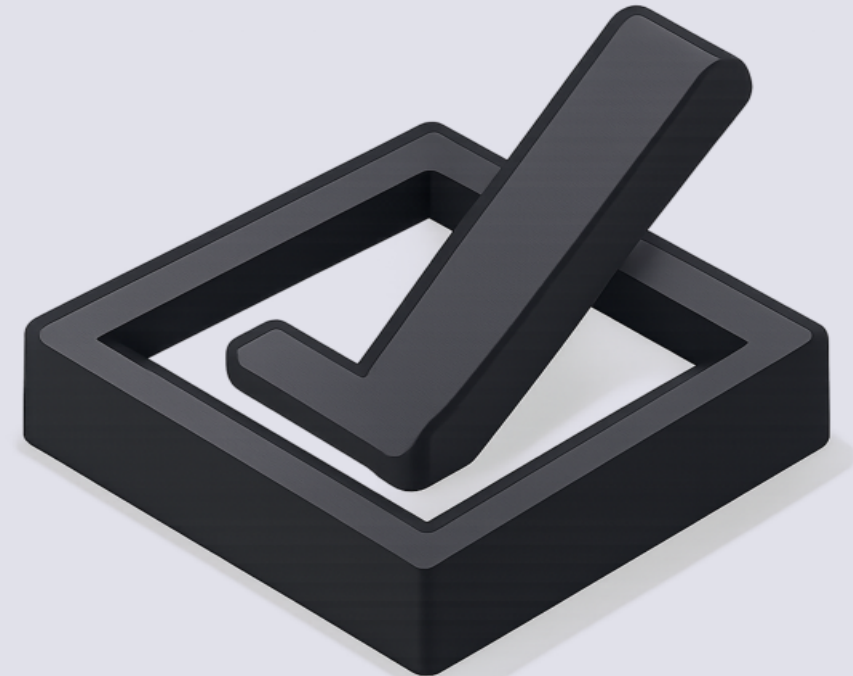
5+ suspicious keywords detected. Displayed in red for a high likelihood of misleading or fabricated content.



Results & Discussion

- ✓ Successfully accepts news article input from users
- ✓ Successfully detects predefined suspicious keywords in the text
- ✓ Correctly counts matching keywords
- ✓ Displays three levels of classification (Possibly Real, Suspicious, Fake)
- ✓ GUI responds correctly to CHECK NEWS and CLEAR buttons
- ✓ System provides a simple preliminary screening mechanism for news articles
- ✓ Demonstrates practical use of Python functions, loops, conditionals, and GUI programming

The system works as intended for a rule-based keyword detection prototype.



Challenges & Limitations

Keyword-based detection cannot understand the actual truth of a news article

A legitimate article may contain suspicious words and produce a false positive

Fake news without predefined keywords may be missed (false negative)

The system depends on the quality and coverage of the keyword list

It does not perform fact-checking against external sources

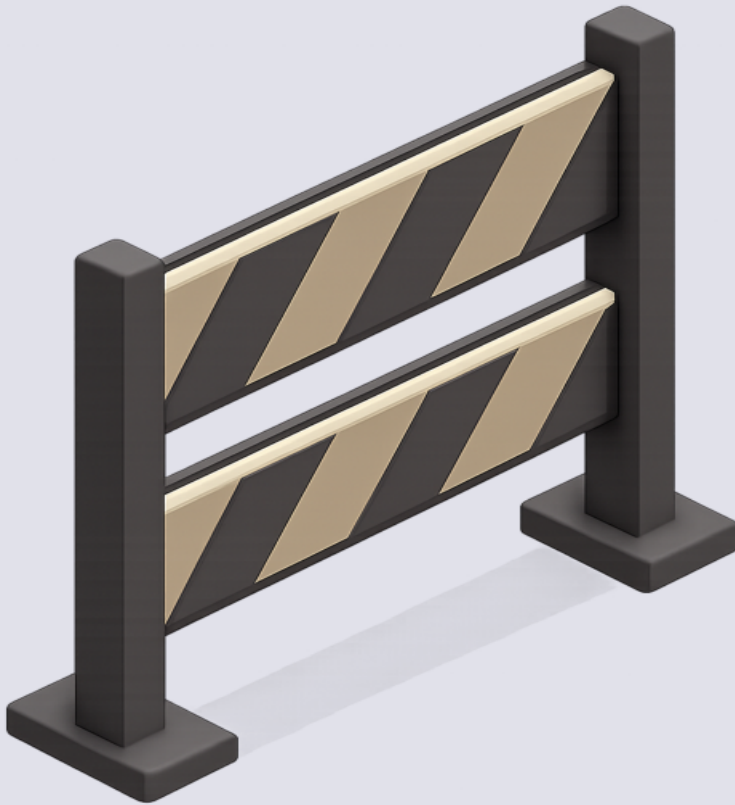
It is a basic rule-based prototype rather than a machine-learning model

No context understanding — only surface-level keyword matching

Keyword list requires manual updates as new phrases emerge

Key Insight

These limitations come from a rule-based approach: the system can match words, but it cannot verify meaning, context, or real-world truth. Future versions can reduce these gaps by adding machine learning, fact-checking APIs, and a continuously updated dataset or keyword library.



The following enhancements can build upon this foundation:

Future Scope

- 1 Use a real labeled fake-news dataset for training
- 2 Apply text preprocessing and feature extraction such as TF-IDF
- 3 Implement machine-learning algorithms such as Random Forest or Logistic Regression
- 4 Compare multiple ML models for accuracy improvement
- 5 Add a confidence score to each classification
- 6 Add a database for storing analysis history
- 7 Integrate online fact-checking APIs
- 8 Improve the GUI with charts and statistics visualization
- 9 Add multi-language support
- 10 Implement sentiment analysis for additional context

Conclusion & Thank You

This project successfully demonstrates how **beginner Python concepts** — functions, loops, conditionals, string operations, lists, and GUI programming — can be combined to build a **practical, real-world application**.

What We Built

A fully functional desktop GUI that detects suspicious keywords and classifies news as Possibly Real, Suspicious, or Fake.

What We Learned

Rule-based systems are transparent and explainable. Understanding their limits is the first step toward more advanced ML solutions.

What Comes Next

Machine learning integration with scikit-learn, live API fact-checking, and multi-language keyword expansion.

Thank You!

Questions and feedback are welcome.



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THANK YOU